ALI ZINDARI

Email ◇ Website ◇ Scholar ◇ LinkedIn ◇ GitHub

RESEARCH INTERESTS

Theory of Deep Learning, Memory

EDUCATION

Ph.D. in Computer Science (ELLIS Program) *Nov. 2025 – Present*
CISPA and EPFL Saarbrücken, Germany
Co-supervised by Prof. Sebastian U. Stich and Prof. Martin Jaggi.

M.Sc. in Mathematics & Computer Science *2023 – 2025*
Saarland University Saarbrücken, Germany
Thesis: Fine-Grained Analysis of Local SGD under Intermittent Communication.

B.Sc. in Computer Engineering *2017 – 2022*
Isfahan University of Technology Isfahan, Iran
Thesis: Segmentation of Lungs COVID Infected Regions by Attention Mechanism and Synthetic Data.

PUBLICATIONS

Learning When to Adapt
A.Zindari, X.Jiang, R.Mulayoff, S.U.Stich — arXiv 2026

LoRA vs. Full Fine-Tuning: A Theoretical Perspective
A.Zindari*, R.Mulayoff*, S.U.Stich — arXiv 2026

Revisiting Consensus Error: A Fine-grained Analysis of Local SGD under Second-order Data Heterogeneity
K.K.Patel, A.Zindari, S.U.Stich, L.Wang — NeurIPS 2025

Decoupled SGDA for Games with Intermittent Strategy Communication
A.Zindari*, P.Yazdkhasti*, A.Rodomanov, T.Chavdarova, S.U.Stich — ICML 2025

The Limits and Potentials of Local SGD for Distributed Heterogeneous Learning with Intermittent Communication
K.K.Patel, M.Glasgow, A.Zindari, L.Wang, S.U.Stich, Z.Cheng, N.Joshi, N.Srebro — COLT 2024

On the Convergence of Local SGD Under Third-Order Smoothness and Hessian Similarity
A.Zindari, R.Luo, S.U.Stich — Opt4ML @ NeurIPS 2023

Segmentation of Lungs COVID Infected Regions by Attention Mechanism and Synthetic Generated Data
A.Zindari*, P.Yazdkhasti*, Z.Nabizadeh, P.Khadivi, N.Karimi, S.Samavi — arXiv 2021

Bifurcated Autoencoder for Segmentation of COVID-19 Infected Regions in CT Image
P.Yazdkhasti*, A.Zindari*, Z.Nabizadeh, R.Roshandel, P.Khadivi, N.Karimi, S.Samavi — ICPR 2021 Workshops

* Equal contribution

RESEARCH EXPERIENCE

CISPA Helmholtz Center for Information Security - MLO Lab Oct. 2022 – Nov. 2025
***Supervisors**: Prof. Sebastian U. Stich - Prof. Tatjana Chavdarova* *Saarbrücken, Germany*

· Worked on theory of distributed optimization and minimax games.

École polytechnique fédérale de Lausanne (EPFL) - LIONS Lab Jun. 2022 – Oct. 2022
***Supervisors**: Prof. Ali Ramezani-Kebrya - Prof. Reza Shokri (NUS)* *Lausanne, Switzerland*

· Worked on the adversarial robustness of self-supervised models.

École polytechnique fédérale de Lausanne (EPFL) - VITA Lab Jul. 2021 – Oct. 2021
***Supervisors**: Prof. Alexandre Alahi - Yuejiang Liu - Mohammadhossein Bahari* *Lausanne, Switzerland*

· Worked on the Motion Forecasting problem for self-driving cars.

Isfahan University of Technology - Biomedical Imaging Lab Jun. 2020 – Feb. 2021
***Supervisors**: Prof. Shadrokh Samavi - Prof. Nader Karimi* *Isfahan, Iran*

· Proposed a bifurcated neural network for segmentation of COVID-19 infected regions in CT images.